17 VESSEL COMPONENT INPUT DATA

This chapter describes input data card requirements for the problems allowed in the 3-D Vessel part of MARS code.

17.1 CARD 60000000, Main Problem Control Data

The three words can be entered on this card.

W1(I)	init	Enter the vessel initialization option. Valid entries are:
		0 = initial start with automatic initialization of the 3D module.
		1 = initial start without automatic initialization of the 3D module.
		2 = initial start without automatic initialization of the 3D module.
		Card 61600NNN should be entered, which are used to specify initial void fractions of the 3D cells.
		4 = fill vessel arrays with data obtained from a restart file
W2(I)	dstep	Enter the time step number of the dump to be used for restarting.
		Enter zero(0) if this is not a restart.
W3(R)	timet	Enter the restart time for the problem.
		Enter zero (0.0) if this is not a restart.

17.2 CARDS 60000101 THROUGH 60000199, Plot File Requests

These cards are optional for NEW and RESTART problems, are required for a REEDIT problem, and are not allowed for PLOT and STRIP problems. If these cards are not present, no minor edits are printed. If these cards are present, minor edits are generated, and the order of the printed quantities is given by the card number of the request card. One request is entered per card, and the card numbers need not be consecutive. For RESTART problems, if these cards are entered, all the cards from the previous problem are deleted.

W1(A)	Variable code (alphanumeric).
W2(I)	Parameter (numeric).

W3(R) Minimum Y Value for X-Y Plot (Optional Select for Window Plotting).

W4(R) Maximum Y Value for X-Y Plot (Optional Select for Window Plotting).

W5(I) Window Identification for multiple X-Y Plot.

W6(I) Color Pallet Identification for multiple X-Y Plot.

Words 1 and 2 form the variable request code pair. The quantities that can be edited and the input required are listed below. For convenience, quantities that can be used in plotting requests, in trip specifications, as search variables in tables, and as operands in control statements are listed. Units for the quantities are also given. Interactive input variables described in Section 6 can be used in batch or interactive jobs in the same manner as the variables listed below. The parameter for interactive input variables is 1000000000. Quantities compared in variable trips must have the same units, and input to tables specified by variable request codes must have the specified units. The quantities are listed in alphabetical order within each section.

17.2.1 Channel/Node Quantities

The quantities listed below are unique to certain components; for example, three-digit number ccc used for channel number and xx for node number in the input cards (for cccxx0000 numeric field)

AE Entrained liquid volume fraction.

AL Vapor volume fraction.

ALIQ Liquid volume fraction.

FEM Vertical entrained liquid momentum flow.

FGM Vertical vapor momentum flow.

FLM Vertical liquid momentum flow.

GAMA Vapor generation rate

P Old time pressure

HL Liquid enthaphy.

HV Vapor enthaphy.

RL Liquid density.

RV	Vapor density.
PMGAS	Gas partial pressure.
RMGAS	Gas density.
QCHFF	Cell critical heat flux
QLIQ	Heat transfer rate to liquid
QVAP	Heat transfer rate to vapor
TEMP_F	Liquid temperature
TEMP_G	Vapor temperature
QUALF	Flow quality

17.2.2 Heat Rod Quantities

The quantities listed below are unique to certain components; for example, three-digit number mmm used for rod number, and 0nn for node number from the bottom of 3D model, and rr for radial mesh number in input card (for mmm0nnrr numeric field) If radial mesh is not available, then use final 2 digit as null number, i.e. mmm0nn00.

TROD	Rod mesh temperature
XC	Axial node elevation for each rod.
QROD	Heat Flux for each rod at certain node.
HTCL	Heat Transfer Coefficient to the liquid
HTCV	Heat transfer coefficient to vapor
TLIQ	Liquid temperature seen by rod
TVAP	Vapor temperature seen by rod
TFAVG	Fuel temperature averaged radially over pellet and axially over fluid node.

17.2.3 Channel Quantities

The quantities listed as unique to a certain component; for example, three digit number ccc used for channel number:

LVL_LIQ Collapsed liquid level for a channel

17.2.4 Gap Qualtities

WGM Vapor mass flow rate in transverse momentum cell

WEM Entrained liquid mass flow rate in transverse momentum cell

WLM Liquid mass flow rate in transverse momentum cell

17.3 Calculation Variables and Initial Conditions

In case of a 'restart problem', following data cards are skipped.

17.3.1 CARD 60100000, Fluid Initial Condition Card

The seven words have to be entered on this card.

W1(R) pref Enter the initial 3D region operating pressure (psi or N/m^2).

W2(R) hin Enter the temperature for fluid initialization (^{o}F or K).

W3(R) gin Enter the vertical mass velocity for flow initialization.(lbm/ft^{2}sec or kg/m^{2}sec).

W4(R) aflux Enter the average linear heat rate per active rod (kW/ft or kW/m).

W5(R) vfrac(1) Volume fraction of liquid

W6(R) vfrac(2) Volume fraction of vapor in gas mixture

W7(R) rbsf Rod bundle scaling factor. Enter 1.0 for ideal subchannel

17.3.2 CARD 60110000, Noncondensible Gas

W1(I) ngas Number of noncondensible gases (minimum of one).

17.3.3 CARD 60110001 through 60110099, Noncondensible Gas Fraction

The two words have to be entered on each card.

W1(A) gtype(i) Enter name of gas (left justified).

Examples: air, argon, helium, hydro, kryto, nitro, oxyge, xneno.

W2(R) vfrac(i+2) Volume fraction of gtype(i) in gas mixture.

17.4 CARD 602xxxxx, Channel Description

17.4.1 CARD 60200000

W1(I) nchanl Enter the number of channels in the problem.

17.4.2 CARD 6020NN00, Channel Geometric Data(NN=1, W1 of 60200000)

The six words have to be entered on each card. The seventh and eighth words may not be entered. NN does not need to be consecutive, however, has to be the same number as the NN of 17.4.1.

W1(I) i Enter the channel identification number.

(Note : Channel index numbers must be unique, but they do not have to be sequential Skipping numbers is permitted, so long as exactly nchanl of 60200000 channels are identified.)

W2(R) an(i) Enter the nominal channel area (in^2 or m^2). (Do not enter zero.)

W3(R) pw(i) Enter the channel wetted perimeter (inches or m). (Do not enter zero.)

W4(R) abot(i) Enter the area of the bottom of the channel for use in the momentum equation. Units are in^2 or m^2. If abot(i)is entered as zero (0.0), it is set to an(i).

W5(R) atop(i) Enter the area on the top of the channel for use in the momentum equation. Units are in^2 or m^2. If atop(i) is entered as zero (0.0), it is set to an(i).

W6(I) namgap Enter the number of gaps for which the vertical velocity of channel i convects transverse momentum between sections.

W7(R) rufnes Enter wall roughness for the wall friction of channel i or gaps connected with the channel i.

W8(R) dhydn Enter hydraulic diameter of the channel (inches or m). dhydn is always greater than 0.0.

17.4.3 CARD 6020NNMM, Channel Geometric Data(MM=1,namgap of 6020NN00)

The three words have to be entered on each card. MM did not need to be consecutive, however, NN has to be the same as the NN of 17.4.1.

NOTE : This card is read only namgap of 6020NN00 > 0. Omit this card if namgap of 6020NN00 is zero (0).

W1(I) inode(i,mm) Enter the index number of the node where the vertical velocity of channel described in 6020NN00 convects transverse momentum across a section boundary.

(Note : inode will be either at the bottom of the channel inode=1, or the top of the channel, inode=nonode+1), where nonode is the number of axial levels in the section containing channel 6020NN00. inode is defined in the section where the vertical momentum equation is solved.)

W2(I) kgapb(i,mm) Enter the index number of the gap below the section boundary. Enter zero if there is no gap below the section boundary.

(Note : If kgapb is not zero, the positive velocity of channel 6020NN00 at inode convects transverse momentum out of kgapb into kgapa. The negative velocity of channel i at inode convects transverse momentum from kgapa into kgapb; but if kgapb is zero, this momentum is dissipated.)

W3(I) kgapa(i,mm) Enter the index number of the gap above the section boundary. Enter zero if there is no gap above the section boundary.

(Note : If kgapa is not zero, the positive velocity of channel i at inode convects transverse momentum from kgapb (if kgapb is not equal 0) into kgapa. If kgapa is zero, this momentum is dissipated. The negative velocity of channel i at inode convects transverse momentum from kgapa to kgapb, if kgapb is not equal 0; if kgapb is zero, this momentum is dissipated.)

17.5 CARD 603XXXXX, Transverse Channel Connection (Gap) Data

These cards are omitted if there are no transverse connections between channels.

17.5.1 CARD 60300000

W1(I) nk Enter the number of transverse connections (gaps).

17.5.2 CARD 603NNN00 through 603NNN99 (NNN=1,nk < 200)

The eighteen words have to be entered on one or more card. NNN does not need to be consecutive.

W1(I)	k	Enter the gap identification number. (Note: Gap numbers must be unique but they do not have to be sequential. nk gaps must be input.)
W2(I)	ik(k)	Enter the identification number of the lower-numbered channel of the pair that connects through gap k.
W3(I)	jk(k)	Enter the identification number of the higher-numbered channel of the pair that connects through gap k.
W4(R)	gapn(k)	Enter the nominal gap width (in. or m).
W5(R)	length(k)	Enter the distance between the center of channel ik(k) and the center of channel jk(k) (in. or m).
W6(R)	wkr(k)	Enter the loss coefficient (velocity head) for gap k.
W7(R)	fwall(k)	Enter the wall friction factor for the gap. Valid entries are : 0.0 = no walls, 0.5 = one wall, 1.0 = two walls
W8(I)	igapb(k)	Enter the index number of the gap below gap k. Enter zero if there is no gap below gap k. (NOTE: The velocity of igapb(k) convects vertical momentum at node i into (or out of) channel ik(k) out of (or into) jk(k).)
W9(I)	igapa(k)	Enter the index number of the gap above gap k. Enter zero if there is no gap above gap k. (NOTE: The velocity of igapa(k) convects vertical momentum at the top node of the section into (or out of) channel ik(k) out of (or into) jk(k).)
W10(R)	factor(k)	Enter 1.0 if gap positive flow (from channel ik(k) to channel jk(k)) is in the same direction as positive flow for the global coordinate system. Enter -1.0 if gap positive flow is opposite to positive flow for the global coordinate system. (Default=1.0).
W11(I)	igap(k,1)	Enter the index numbers of gaps facing the ik(k) side of gap k. If the gap faces a wall, enter -1.
W12(I)	jgap(k,1)	Enter the index numbers of gaps facing the jk(k) side of gap k. If the gap faces a wall, enter -1.
W13(I)	igap(k,2)	
W14(I)	jgap(k,2)	
W15(I)	igap(k,3)	
W16(I)	jgap(k,3)	

Three sets of (W13(igap)~W14(jgap)) and (W15(igap)~W16(jgap)) have to be entered.

Note : The input for W10~W16 is required only if the three-dimensional form of the transverse momentum equation is desired.

W17(R) gmult(k) Enter the number of actual gaps modeled by gap k.

W18(R) etanr(k) Enter the crossflow de-entrainment fraction.

17.6 CARD 60320000

W1(I) nlmgap Enter the number of gaps that convect orthogonal transverse momentum. (This is required only for the three-dimensional form of the transverse momentum equation.) Enter zero if the three-dimensional form of the transverse momentum equation is not desired.

17.6.1 CARD 60320100 through 60329900

Read only if nlmgap of 60320000 > 0. These card numbers have to be consecutive.

W1(I) kgap1(n) Enter the index number of a gap whose velocity transports transverse momentum from one gap to another.

W2(I) kgap2(n) Enter the index number of the gap that receives the transverse momentum convected by the positive velocity of gap kgap1. A nonzero value must be entered.

W3(I) kgap3(n) Enter the index number of the gap that the positive velocity of kgap1 transports the transverse momentum out of. A nonzero number must be entered.

(Note : The positive velocity of kgap1 transports momentum from kgap3 to kgap2. The negative velocity of kgap1 will transport transverse momentum in the opposite direction; i.e., from kgap2 into kgap3.)

17.7 CARD 604XXXXX, Vertical Channel Connection Data

17.7.1 CARD 60400000

A word has to be entered on this card.

W1(I) nsects Enter the number of sections in this problem.

17.7.2 CARD 604K0000, 604K00MM, 604KNN00, 604KNN01, 604KNN11 (K=1, nsects)

The K does not need to be consecutive, however, all cards of the 604K0000, 604K00MM, and 604KNN00 must have the same K.

17.7.2.1 CARD 604K0000

The five words have to be entered on each card.

W1(I)	isec	Enter the section identification number. If this number is positive, Cards 604KNN00 (old format) should be entered. If this number is negative, Cards 604KNN01 and 604KNN11 (new format) should be entered. And WI(I) of other 604K0000 cards should be negative.
W2(I)	nchn	Enter the number of channels in section isec.
W3(I)	nonode	Enter the number of vertical levels in section isec.
W4(R)	dxs(isec,1)	Enter the vertical node length in this section (in.or m).
W5(I)	ivardx	Flag for variable node length in this section. For constant node length, this word is 0(default). If this word is greater than 0, DX table is read ivardx pairs in variable.

17.7.2.2 CARD 604K00MM (MM=1,ivardx)

The two words have to be entered on each card.

W1(I)	jlev(i)	Last axial level in section to have a node length of vardx(i) of this card. jlev(ivardx) of the last card (604k00MM,MM=W5 of 604K0000) must be greater than or equal to nonode+1.
W2(R)	vardx(i)	Axial node length (in. or m).

17.7.2.3 CARD 604KNN00 (NN=1,nchn)

These cards have to be entered only when W1 of Card 604K0000 is positive. The thirteen words have to be entered on each card.

W1(I)	i	Enter the identification number of a channel in section isec.
W2~7(I)	kchana(i,j)	Enter the indices of channels in the section above isec that connect to channel i of this card. If channel i of this card does not connect to any channels above, enter i of this card in W2(I).

W8~13(I) kchanb(i,j) Enter the indices of channels in the section below isec that connect to channel i of this card. If channel W1 of this card does not connect to any channels below, enter i of this card in W9(I).

17.7.2.4 CARD 604KNN01 (NN=1,nchn)

These cards have to be entered only when W1 of Card 604K0000 is negative. Forty one words may be entered on each card.

W1(I) i Enter the identification number of a channel in section isec.

W2~41(I) kchana(i,j) Enter the indices of channels in the section above isec that connect to channel i of this card. If channel i of this card does not connect to any channels above, enter i of this card in W2(I).

17.7.2.5 CARD 604KNN11 (NN=1,nchn)

These cards have to be entered only when W1 of Card 604K0000 is negative. Forty one words may be entered on each card.

W1(I) i Enter the identification number of a channel in section isec.

W2~41(I) kchanb(i,j) Enter the indices of channels in the section below isec that connect to channel i of this card. If channel W1 of this card does not connect to any channels below, enter i of this card in W2(I).

17.8 CARD 605XXXXX, Geometry Variation Data

The input for these cards allows the user to specify vertical variations in the continuity area, momentum area, or wetted perimeter for channels, and in the transverse width for gaps. It can be omitted if such variations are not needed.

17.8.1 CARD 60500000

W1(I) nafact Enter the number of geometry variation tables to be entered.

17.8.2 CARD 6050NN00, 6050NNMM

These cards are repeated nafact times

17.8.2.1 CARD 6050NN00 (NN=1, nafact)

The NN has to be consecutive.

W1(I) nxl(nn) Enter the number of points in this variation table.

17.8.2.2 CARD 6050NNMM (MM=1, nxl(nn))

The MM has to be consecutive.

W1(I) jaxl(nn,mm) Enter the node number at which to apply the area variation factor for table NN, point MM.

W2(R) afact(nn,mm) Enter the variation factor for table NN, point MM.

Area = afact(nn,mm) X an(i), or

Gap width = afact(nn,mm) X gapn(k)

Gap loss coefficient = afact(nn,mm) X wkr(k)

17.9 CARD 606XXXXX, Channels and Gaps Affected by Variation Tables

17.9.1 CARD 60600000

W1(I) n1 Enter the total number of channel and gap variation table cards to be read.

17.9.2 CARD 6060NN00 through 6060NN99 (NN=1,n1)

The fifteen word have to be entered on one or more cards.

W1(I) iact Enter a positive integer corresponding to a variation table number, for channel continuity area variation. Enter a negative integer, whose absolute value corresponds to a variation table number, for gap width variation. Enter zero (0) if iamt is negative.

W2(I) iamt Enter a variation table number for channel momentum area variation. Enter a negative integer, whose absolute value corresponds to a variation table number, for gap loss coefficient variation. Enter zero (0) if iact is negative.

W3(I) ipwt Enter a variation table number for wetted perimeter variation. Enter zero (0) if iact or iamt is negative.

W4~15(I) icrg(m) Enter the index numbers of the channels (or gaps if iact(iamt) is negative) that the tables identified in iact, iamt and ipwt are to be applied to.

17.10 CARD 607XXXXX, Local Loss Coefficient and Grid Spacer Data.

This card can be omitted if they are not needed.

17.10.1 CARD 60700000

The eight words have to be entered on this card

W1(I) ncd Enter the number of loss coefficient specifications to be read. (These include vertical momentum losses only. Transverse losses are specified in 603NNN00)

W2(I) ngt Number of grid types to be read.

W3(I) ifgqf Flag for grid quench front model (1=on, 0=off).

W4(I) ifsdrp Flag for small drop model (1=on, 0=off).

W5(I) ifespv Flag for grid convective enhancement (1=on, 0=off).

W6(I) iftpe Flag for two-phase enhancement of dispersed flow heat transfer (1=on, 0=off).

W7(I) igtemp Number of the sets of grid temperature corresponding to grid elevation

W8(I) nfbs Number of flow blockages

17.10.2 CARD 607001NN (NN=1,ncd), Loss Coefficient Data

The fourteen words can be entered on each card. This card is repeated W1 of 60700000 times. The card numbers have to be consecutive.

W1(R) cdl Enter the loss coefficient (velocity head).

W2(I) j Enter the node number where the loss coefficient is applied. (NOTE: The vertical node number is relative to the beginning of the section containing the channel(s) listed in icdum(i).)

W3~14(I) icdum(i) Enter the index number(s) of channel(s) the loss coefficient will be applied to at node j. (Up to twelve channels may use the specified loss coefficient cdl at vertical node j.)

17.10.3 CARD 607002NN(NN=1,igtemp), Grid Temperature Data(UNIDENTIFIED)

The two words can be entered on each card. This is repeated in igtemp times.

W1(R)	tgrid	Enter the grid temperature at elevation W2. (^{o}F or K)
W2(R)	qfgrid	Enter the grid elevation (in. or m).

17.10.4 CARD 6070KXXX (K=1,W8 of 60700000), Flow Blockages Data

17.10.4.1 CARD 6070K000

The ten words have to be entered on this card. This card is repeated W8 of 60700000 times.

W1(I)	i	Flow blockage index number
W2(I)	ifb	Channel index number
W3(I)	jsfb	Axial node index number
W4(I)	nrfb	Number of rods that block this channel
W5(R)	spoint	Axial position of the flow separation point.
W6(R)	dsep	Channel diameter at separation point,
W7(R)	throat	Diffuser diameter at exit
W8(R)	aflblk	Area for DBM (fraction of channel) x 0.25
W9(R)	cdfb	Loss coefficient (Rehme multiplier)
W10(R)	ablock	Blockage area ratio

17.10.4.2 CARD 6070K00N (Nn=1, W4 of 6070K000(nrfb))

W1(I)	nrodfb(n)	Index number of rod.
W2(I)	krodfb(n)	Surface number.
W3(R)	angiht(n)	Angle for impact heat transfer (degree).
W4(R)	araiht(n)	Area for impact heat transfer(in^2/per rod).

17.10.5 CARD 607KXXXX (k=1,ngt), Grid Type Data

607k0000, 607k00xx, and 607knn00 have to be specified ngt times

17.10.5.1 CARD 607K0000

The eight words have to be entered on each card.

W1(I)	ing	Grid type number. (must be sequential starting with 1)
W2(I)	ngal(ing)	Number of axial locations for grid type ing.
W3(I)	ngcl(ing)	Number of channels containing grid ing at levels ngal(ing).
W4(I)	igmat(ing)	Grid material table index (MMM) . Refer 1-D material table (201MMMNN).
W5(R)	gloss(ing)	Loss coefficient multiplier (suggest 1.0 for round edge grids; 1.4 for square edge grids)
W6(R)	gabloc(ing)	Fraction of channel area blocked by grid
W7(R)	glong(ing)	Grid length (in. or m)
W8(R)	gperim(ing)	Grid perimeter (in. or m)

17.10.5.2 CARD 607K0001 througth 607K0099

The number of words are equal to ngal(ing)

W1~NN(I)	nngl(ing)	Axial node number of momentum cells containing grid type ing.

17.10.5.3 CARD 607KNN00 (nn=1, ngcl(ing))

The fourteen words have to be entered on each card. This card is repeated ngcl(ing) times per ing.

W1(I)	ncngl(ing,nn)	Channel ID number with grid type ing at axial levels nngl(ing) (specified above).
W2(R)	gmul(ing,nn)	Number of grids contained in channel ncngl(ing,nn).
W3,5,7,(I)	ngrod(*)	Whole rod number with surface surrounding grid(maximum of six)
W4,6,8,(I)	ngsurf(*)	Rod surface index of whole rod ngrod surrounding grid ing.
		* : 3-dimensional variable, (ing, nn, m). Here m =1,6.

(Note: average temperature of all surfaces surrounding grid is used to transport heat between grid and heater rods.)

17.11 CARD 608XXXXX, Rod and Unheated Conductor Data

This card can be omitted if they are not needed.

17.11.1 CARD 60800000

The eight words can be entered on this card. Rods and unheated conductors are both used to model solid coducting structures in vessel. Rods can model either active or passive element, but unheated conductors are always passive. Unheated conductors can not have internal heat sources.

W1(I) nrod Enter number of active or passive conducting rods, including fuel rods.

W2(I) nsrod Enter number of unheated conductors.

W3(I) nc Conduction model flag;

nc =1 : for radial conduction only.

nc =2 : for radial and axial conduction.

nc =3 : for radial, axial, and azimuthal conduction.

W4(I) nrtab Enter number of temperature initialization tables to be read.

W5(I) nrad Enter number of radiation channels.

W6(I) nltyp Enter number of location types.

W7(I) nsrad Enter total number of rod or slab surfaces with radiation.

W8(I) nxf Enter number of time steps between radiation calculations (Default=1).

17.11.2 CARD 6080NN00 (NN=1,nrod), Rod Geometry Data

The cards in 17.11.2 is specified to define the geometry of structures that generate heat, including nuclear fuel rods. It is repeated nrod times. The eleven words can be entered on each card.

W1(I) n Enter rod identification number. (Note: Rod index numbers must be entered sequentially, from 1 to W1 of 60800000. Skipping numbers is not permitted.)

W2(I) iftyp(n) Enter geometry type identification number. (Refer to 609xxxxx card for geometry type input data.)

W3(I) iaxp(n) Enter axial power profile table identification number. (Refer to 611xxxxx card for axial power profile input data.) If 0 is entered, the source of the heat structure is taken from the 3D kinetics module, MASTER. In this case, Input Cards 61100001 and 61100002 should be provided. In addition, MASTER input files and mapping data ("MAS_MAP" file) should be also prepared. [see Appendix A, or KAERI/TR-2232/2002]. If a negative integer is entered, the source of the heat structure is taken from the nuclear rod, of which rod number is "-iaxp(n)."

W4(I) nrenode(n) Enter renoding flag for heat transfer solution for rod n.

= 0 : no fine mesh renoding

> 0 : renoding every this word time steps

< 0 : renoding every this word time steps, based on inside surface temperatures

W5(R) daxmin(n) Enter minimum axial node size (in. or m). (This is used only if fine mesh renoding is used.)

W6(R) rmult(n) Enter rod multiplication factor (number of rods modeled by rod n). (This number can contain fractional parts.)

W7(R) radial(n) Enter radial power factor (normalized to average power).

W8(R) hgap(n) Enter constant gap conductance (Btu/hr-ft^2-^{o}F or W/m^2-^{o}C). (This parameter is used only for nuclear fuel rods that do not have the dynamic gap conductance model specified by their geometry type.)

Enter 0 if the rod n does not model a nuclear fuel rod.

W9(I) isecr(n) Number of sections containing rod n.

W10(R) htamb(n) Heat transfer coefficient for heat loss to ambient from surface not connected to a channel (Btu/hr-ft^2-^{o}F or W/m^2-^{o}C).

W11(R) tamb(n) Sink temperature for ambient heat loss (^{o}F or ^{o}K).

17.11.3 CARD 6080NNK0 (K=1,isecr(n))

The eight words can be entered on each card. This card is repeated isecr(n).

W1(I) nischc(n,k) Negative of channel number connected to the inside of rod n.

17.11.4 CARD 6080NNK1 through 6080NNK9

The sixteen words can be entered on each card. This card is repeated isecr(n).

W1(I) nschc(k) Channel number with thermal connections to rod n. Enter 0 if no channels are connected to the outside of rod n.

W2(R) pie(k) Fraction of rod n thermally connected to channel nschc(k).

Odd words are the same as the Word 1(nschc(k)), and even words are the same as the Word 2 (pie(k)). The eight pairs can be specified.

17.11.5 CARD 6081NN00 (NN=1, nsrod), Unheated Conductor Data

This card is read for each of the nsrod conductor rods (also called heat slabs).

Specify only if nsrod > 0.

W1(I) n Enter unheated conductor identification number. (Note : Unheated conductor index numbers must be entered sequentially, from 1 to nsrod. Skipping numbers is not permitted.)

W2(I) istyp(n) Enter geometry type identification number. (Refer to 609xxxxx card for geometry type input data.)

W3(R) hperim(n) Enter wetted perimeter on outside surface (in. or m).

W4(R) hperimi(n) Enter wetted perimeter on inside surface (in. or m). Enter zero for a solid cylinder.

W5(R) rmuls(n) Enter multiplication factor (number of elements modeled by unheated conductor n).

(This number can contain fractional parts.)

W6(I) nslchc(n) Channel number on inside of slab.

W7(I) ndslch(n) Channel number on outside of slab.

W8(R) htambs(n) Heat transfer coefficient for heat loss to the ambient (Btu/hr-ft^2-^{o}F or W/m^2-^{o}C)

W9(R) tambs(n) Sink temperature for ambient heat loss (^{o}F or K)

17.11.6 CARD 6082XXXX, Rod Temperature Initialization Tables

Cards 6082XXXXX are specified which temperature tables apply to which rods and unheated conductors. The sequence is repeated nrtab times, and all rods and conductors must be accounted for.

17.11.6.1 CARD 6082NN00

The four words have to be entered on each card.

W1(I) i Enter identification number of temperature table.

W2(I) nrt1 Enter number of rods using table i.

W3(I) nst1 Enter number of unheated conductors using table i.

W4(I) nrax1 Enter number of pairs of elements in table i.

In case of a tube inside, use a negative number.

17.11.6.2 CARD 6082NN01 through 6082NN05

Specify only if nrt1 > 0. The number of words are nrt1.

W1,2,3,(I) irtab(i,l) Enter identification numbers of rods using table i for temperature initialization.

Note : The steady-state conduction equation is solved for these rods using the temperatures from this table as a boundary condition on the rod surface.

17.11.6.3 CARD 6082NN06 through 6082NN10

Specify only if nst1 > 0. This card has to contain nst1 words.

W1,2,3,(I) istab(i,l) Enter identification numbers of unheated conductors using table i described in 6082NN00 for temperature initialization.

Note : A flat radial temperature profile is assumed initially in unheated conductors.

17.11.6.4 CARD 6082NN11 through 6082NN99

This card are required if W4 of card 6082NN00(nrax1) is not zero. In format 1(W4 of card 6082NN00 > 0), each data set has two words. In format 2(W4 of card 6082NN00 < 0), each data set has three words. These cards have to contain data sets of | nrax1 |.

Format1

W1(R) axialt(i,l) Enter the vertical position relative to the bottom of the lowest section (in. or m).

W2(R) trinit(i,l) Enter the temperature to be applied at axialt(i,l) (^{o}F or K).

Format2

W1(R) axialt(i,l) Enter the vertical position relative to the bottom of the lowest section (in. or m).

W2(R) trinit(i,l) Enter the temperature to be applied at axialt(i,l) (^{o}F or K). In tube, inner side temperature are specified.

W3(R) tronit(i,l) Enter the temperature to be applied at axialt(i,l) (^{o}F or K). In tube, outer side temperature are specified.

17.11.7 CARD 6083XXXX though 6084XXXX, Radiation Initialization Tables

Cards 6083XXXX through 6084XXXX are read in to specify orientation and which location type tables apply to which fluid channels, rods, and unheated conductors if W5 of 60800000 > 0.

17.11.7.1 CARD 6083NN00(NN=1,nrad), Channel Orientation and Location Type

This card repeated W5 of 60800000(nrad) times. The ten words are entered on each card.

W1(I) idcard(nn) Radiation channel ID number

W2(I) nsidr(nn) Number of fluid channel which contains idcard(nn).

W3(I) locate(nn) Location type for radiation channel idcard(nn) :

<0 : contains no unheated conductors.

>0 : has both rods and unheated conductors.

W4(I) nrrad(nn) Number of contributing radiation surfaces for idcard(nn):

= 20: location types 1,2,3,7,8

= 16: location types 4,5

= 9: location type 6

= 14: location type 9

W5(I)	nsymf(nn)	Enter flag for fluid channel or rod lumping. = 0 : no lumping = 1 : lumped fluid channels
W6(R)	mltf(1,nn)	Enter surface lumping factor for surface position 1. Ratio of total calculated to actually modeled surface areas of this rod type contained in location type idcard(nn) times the ratio of total surface areas in all channels of this rod type to this surface area (Default=1.0).
W7(R)	mltf(2,nn)	Enter surface lumping factor for surface position 2. Ratio of total calculated to actually modeled surface areas of this rod type contained in location type idcard(nn) times the ratio of total surface areas in all channels of this rod type to this surface area (Default=1.0).
W8(R)	mltf(3,nn)	Enter surface lumping factor for surface position 3. Ratio of total calculated to actually modeled surface areas of this rod type contained in location type idcard(nn) times the ratio of total surface areas in all channels of this rod type to this surface area (Default=1.0).
W9(R)	mltf(4,nn)	Enter surface lumping factor for surface position 4. Ratio of total calculated to actually modeled surface areas of this rod type contained in location type idcard(nn) times the ratio of total surface areas in all channels of this rod type to this surface area (Default=1.0).
W10(R)	vdmlt(nn)	Vapor/droplet multiplication factor. Total number of radiation channels being modeled by this location type (Default=1.0).

17.11.7.2 CARD 6083NN01 through 6083NN99, Radiation Channel Orientation Array

The number of words are W4 of 6083NN00(nrrad(nn)). j=1,nrrad(nn)

W1,2,3..(I)	lrad(nn,j)	Rod number in position "j" for appropriate radiation channel idcard(nn). Negative for inside surface. (See text for proper rod orientation.)

17.11.7.3 CARD 6084K000(K=1,W6 of 60800000(nltyp)), Radiation Location Type Information

This card is repeated nltyp times.

W1(I) idtyp(k) Location type to be input

> 0 : manual input to follow.

< 0 : auto view factor routine to be used.

If W1 of 6084K000(idtyp(k)) < 0 skip the manual input cards

17.11.7.4 CARD 6084K101 through 6084K110, Area input, j=1,jtop

The jtop is total number surfaces for location type idtyp(k).

W1,2,..,jtop(R) arad(j) Surface area of position "j" for location type idtyp(k) (in. or centimeter)

= 40 : (idtyp(k) : 1)

= 38 : (idtyp(k) : 2)

= 36 : (idtyp(k) : 3)

= 26 : (idtyp(k) : 4)

= 25 : (idtyp(k) : 5)

= 16 : (idtyp(k) : 6)

= 38 : (idtyp(k) : 7)

= 36 : (idtyp(k) : 8)

= 23 : (idtyp(k) : 9)

17.11.7.5 CARD 6084K111 through 6084K120, Emissivity input, j=1,jtop

The jtop is total number surfaces for location type idtyp(k).

W1,2,..,jtopt(R) erad(j) Enter the emissivity of position "j" for location type idtyp(k)

17.11.7.6 CARD 6084K121 through 6084K380, View Factor input, m=j,jl

jl = Total number of radiant surfaces in location type idtyp(k).

This card number is increased by 20 from 6084K121 after entering a series of data set.

W1,2,(R) frad(j,m) Enter radiation view factor between one surface and another surface.

Ex.) 6084K121 through 608K140, W11,W12,W13,...,W1N

6084K141 through 608K160, W22,W23,...,W2N

6084K161 through 608K180, W33,...,W3N

6084KXX1 through 608KXX0, ...,WNN.

Where Wij is view factor between surface "i" and surface "j". Subscript N is total number of radiant surfaces in location type W1 6084K000(idtyp(k)) :

= 12: (idtyp(k) : 1, 2, 3)

= 9 : (idtyp(k) :4, 5)

= 6 : (idtyp(k) :6)

= 12: (idtyp(k) :7, 8)

= 7 : (idtyp(k) :9)

17.11.7.7 CARD 6084K421 through 6084K680, Beam Length input

This card number is increased by 20 from 6084K421 after entering a series of data set.

W1,2,(R) drad(j,m) Enter radiation beam length between one surface and another surface (in. or cm).

These input formats are the same as view factor input except for the card number.

17.11.7.8 CARD 6084k701, Auto View Factor input

Omit if W1 of 6084K000(idtyp(k)) > 0.

W1(R) apar(1) Enter first parameter for auto view factor input according to location type.

	(d)	Enter nominal rod diameter(inches or centimeters).
W2(R)	apar(2)	Enter second parameter for auto view factor input according to location type.
	(em1)	Enter emissivity of rod(inches or centimeters).
W3(R)	apar(3)	Enter third parameter for auto view factor input.
	(d1)	If location type=1, enter rod diameter of rod in oversized rod location.
	(g)	Enter gap width between rod and wall (inches or centimeters).
	(cld)	If location type=4,5 and W5>0, Enter position of centerline (unidentified)
		If location type=8,9 and $W5 \neq 0$, Enter position of centerline (unidentified)
	(disx)	If location type=6 and $W5 \neq 0$, Enter displacement from centerline axis to rod position 1 (inches or centimeters)
W4(R)	apar(4)	Enter fourth auto view factor input parameter.
	(p)	Enter pitch of rods.
W5(R)	apar(5)	Enter fifth auto view factor input parameter.
	(rad)	Enter radius of curvature of wall.
W6(R)	apar(6)	Enter sixth auto view factor input parameter.
	(disx)	Enter displacement from centerline axis to rod position 1(inches or centimeters)
W7(R)	apar(7)	Enter seventh auto view factor input parameter.
	(em2)	Enter emissivity of the wall. If location type=4,5, enter 0.0
W8(R)	apar(8)	Enter eighth auto view factor input parameter.
	(d1)	If location type=1,2,3, enter rod diameter of rod in oversized rod location. Enter emissivity of the wall.
	(fg)	If location type=7,8,9, enter fg.

If location type=4,5,6, Enter 0.0

W9(R) apar(9) Enter nineth auto view factor input parameter.

(d1) If location type=7, enter rod diameter of rod in oversized rod location. enter emissivity of the wall.

(fw) If location type=8,9, enter fg. If location type=1,2,3,4,5,6, enter 0.0

17.12 CARD 609XXXXX, Conductor Geometry Description

The geometry types are read in these cards. The geometry types are numbered sequentially in the order they are read in. Nuclear rod geometry types are read using cards 609nn000 through 609nn3xx. All other geometry types are read using cards 609nn000 and 60nn4xx.

17.12.1 CARD 60900000

The four words can be entered on this card.

W1(I) nfuelt Enter number of geometry types to be read in. Note: A geometry type may be used by both rods and unheated conductors, but for the unheated conductor, any heat generation specified for the type will be ignored.

W2(I) irelf Fuel relocation flag (1=on, 0=off) (This is used only for nuclear fuel rods using the dynamic gap conductance model).

W3(I) iconf Fuel degradation flag (1=on, 0=off) (NOTE : if irelf = 1, then iconf = 1.)

W4(I) imwr Flag for metal-water reaction. (zirconium dioxide only)

= 0, off

= 1, Cathcart (for best-estimate analysis)

= 2, Baker-Just (for evaluation model analysis)

17.12.2 CARD 60900NN0

W1(I) i Enter the geometry type identification number. (Note : Geometry type index numbers must be entered sequentially, from 1 to W1 of 60900000(nfuelt). Skipping numbers is not permitted.)

W2(A) ftype(i) Enter the one of the following four words.

nucl : Nuclear fuel geometry

hrod : Solid cylinder

tube : Hollow tube

wall : Flat plate.

The following cards are divided into FORMAT1 and FORMAT2 according to geometry type. Format1 is the information for the nuclear fuel geometry(nucl) geometry type. Format2 is the information for the other geometry types (hrod, tube, wall).

FORMAT 1 : 609NN000 through 609NN3XX (W2 of 60900NN0(ftype(i)) = nucl)

17.12.2.1 CARD 609NN000 through 609NN009

The twelve words have to be entered on each card. Specify only if ftype(i) =nucl.

W1(R) drod(i) Enter rod outside diameter (in. or m)

W2(R) dfuelt(i) Enter fuel pellet diameter (in. or m)

W3(I) nfuel(i) Enter number of radial nodes in fuel pellet.

W4(I) imatf(i) Fuel material properties flag.

Enter zero (0) for built-in U02 properties.

Enter a positive integer corresponding to the identification number of a material properties tables for user-input properties. (Refer 1-D material tables (201MMMNN).)

W5(I) imatc(i) Clad material properties flag :

Enter zero (0) for built-in zirconium properties.

Enter a positive integer corresponding to the identification number of a material properties table for user-input properties. (Refer 1-D material tables (201MMMNN).)

W6(I) imatox(i) Clad oxide property flag:

Enter zero (0) for built-in zirconium dioxide properties.

Enter a positive integer corresponding to the identification number of a material properties table for user-input properties (Refer 1-D material tables (201MMMNN)).

W7(R) dcore(i) Enter diameter of central void for cored fuel (in. or m). Enter zero for uncored fuel.

W8(R) tclad(i) Enter clad thickness (in. or m).

W9(R) ftdens(i) Enter fuel theoretical density as a fraction (used only if built-in U02 properties have been flagged; i.e., if imatf(i) = 0). Note : Do not enter zero.

W10(I) igpc(i) Gap conductance option flag:

Enter zero (0) for constant gap conductance (as specified by W8 on card 6080nn00(hgap(n))).

Enter a positive integer for user-specified non-uniform gap conductance (entered on card 609NN2XX in a table of igpc(i) elements).

Enter a negative integer for the dynamic gap conductance model. (| igpc(i) | is the number of entries in the cold gap width vs axial location table, read on card 609NN2XX)

W11(I) igforc(i) Flag for temporal forcing function on gap conductance (valid only if igpc(i) > 0:

Enter zero (0) for constant gap conductance.

Enter a positive integer for a temporal forcing function with igforc(i) table entries.

W12(I) iradp(i) Enter number of entries in radial power profile table for the fuel pellet.

Enter zero (0) for a uniform radial power profile.

17.12.2.2 CARD 609NN101 through 609NN110

Specify only if W2 of 60900NN0(ftype(i)) = nucl and W10 of 609NN00X(igpc(i)) < 0. The eleven words have to be entered on this card.

W1(R) pgas(i) Enter cold pin gas pressure for nuclear fuel rod geometry type (psia or N/ m^2).

W2(R) vplen(i) Enter gas plenum volume (in^3 or m^3).

W3(R) rouff(i) Enter fuel pellet surface roughness (in. or m).

W4(R) roufc(i) Enter surface roughness of clad inner surface (in. or m).

(Note: Fuel and clad surface roughness should correspond to those used in FRAPCON-2 since the correlation is empirical.)

Fuel surface rouff(i) = 0.000085 inches

Clad surface roufc(i) = 0.000045 inches

W5(R) gsfrac(1) Enter molar fraction of helium gas present.

W6(R) gsfrac(2) Enter molar fraction of xenon gas present.

W7(R) gsfrac(3) Enter molar fraction of argon gas present.

W8(R) gsfrac(4) Enter molar fraction of krypton gas present.

W9(R) gsfrac(5) Enter molar fraction of hydrogen gas present.

W10(R) gsfrac(6) Enter molar fraction of nitrogen gas present.

Note: sum of gas fractions = 1.0

W11(R) oxidet(i) Enter initial oxide thickness for the zircaloy metal-water reaction rate equation(in. or m).

(Used only if W4 of 60900000(imwr) > 0.)

17.12.2.3 CARD 609NN2MM (MM = 1, | W10 of 609NN00X(igpc(i)) |)

Specify only if ftype(i) = nucl and | igpc(i) | > 0. The two words have to be entered on each card.

W1(R) axj(i,mm) Enter topmost vertical position, measured from the bottom of the rod, at which the cold gap width (or gap conductance) agfact(i,mm) is applied. (All vertical levels below axj(i,mm) and above axj(i,mm-1) will have agfact(i,mm) for gap width or gap conductance.) Units on axj(i,mm) are inches or meters.

W2(R) agfact(i,mm) Enter cold gap width if igpc(i) is negative. Units are inches or meters.

Enter gap conductance if igpc(i) is positive. Units are Btu/hr-ft^2-^{o}F or W/m^2-K.

17.12.2.4 CARD 609NN3MM (MM = 1, | W12 of 609NN00X |)

Specify only if W2 of 60900NN0(ftype(i)) = nucl and W12 of 609NN00X(iradp(i)) > 0. The two words have to be entered on each card.

W1(R) radp(mm) Enter the relative radial location (r/ro) where corresponding power factor powr(mm) is applied.

W2(R) powr(mm) Enter the relative power factor (i.e., the ratio of local power at location radp(mm) to total rod power).

FORMAT 2 : 609NN000 and 609NN4XX (W2 of 60900NN0(ftype(i)) = hrod or tube or wall)

17.12.2.5 CARD 609NN000

These data are read for all geometry types that do not describe nuclear fuel. The five words have to be entered on this card. Specify only if W2 of 60900NN0(ftype(i)) ≠ nucl

W1(R) drod(i) Enter outside diameter for hrod or tube geometries (inches or meters).

Enter the wetted perimeter for wall geometries (inches or meters).

W2(R) din(i) Enter inside diameter for tube geometries (inches or meters).

Enter thickness for wall geometries (inches or meters).

Enter zero (0.0) for hrod solid cylinder geometries.

W3(I) nfuel(i) Enter the number of regions within the conductor. (Each region has a uniform power profile and consists of one material.)

W4(I) imatox(i) Enter material property table identification number for oxide on outside surface.

(Refer 1-D material tables (201MMMNN), Default is zirconium oxide; imatox(i) = 0).

Enter the index number of the material property table for material in region nfuel(i) if there is no oxide present.

W5(I) imatix(i) Enter material property table identification number for oxide on inside surface

(Refer 1-D material tables (201MMMNN), Default is zirconium oxide; (imatix(i) = 0); applies only to tube or wall.)

Enter the index number of the material property table for material in region 1 if there is no oxide present.

Data sets for the regions spcified in W3 of 609NN000(nfuel(i)) of geometry type specified in W1 of 60900NN0(i) are entered starting at the centerline for hrod types and at the inside surface for tube and wall types. Data sets are entered in sequence moving radially toward the outside surface.

17.12.2.6 CARD 609NN4MM (MM = 1, | W3 of 609NN000(nfuel(i)) |)

Specify only if W2 of 60900NN0(ftype(i)) ≠ nucl. The four words have to be entered on each card.

W1(I) noder(mm) Enter the number of radial heat transfer nodes in a region.

W2(I) matr(mm) Enter the material property table identification number for the region (Refer 1-D material tables (201MMMNN),.

W3(R) treg(mm) Enter the thickness of the region (inches or meters).

Note : For tube and hrod geometry types,

W4(R) qreg(mm) Enter radial power factor for the region. (This profile is automatically normalized to unity.)

17.13 CARD 611XXXXX, Axial Power Tables and Forcing Functions

These input cards can be omitted if they are not needed. The power forcing functions cards are replaced at restart problem.

17.13.1 CARD 61100000

The one word has to be entered on this card

W1(I) naxp Enter number of axial power profile tables to be read. (Minimum of one.)

17.13.2 CARD 61100001, MARS/MASTER coupling data

If the 1D hydrodynamic module is used:

W1(I) Axial mesh number (NN) of the lower axial reflector region.

W2(I) Axial mesh number (NN) of the upper axial reflector region.

W3(I) Component number (CCCNN) of the bottom volume of the radial reflector region.

W4(R) Starting time of the transient advancement of MASTER (sec).

W5(I) Trip number (reactor trip card number).

W6(R) Total core power (W).

Note: The radial reflector region should be modeled with a "pipe" or "annulus" component. The vertical nodding of the radial reflector should be consistent with that of the active core.

If the 3D hydrodynamic module is used:

W1(I) Axial level for the lower axial reflector region (greater than 1).

W2(I) Axial level for the upper axial reflector region. The lower & upper reflector should be included in the same section. That is, the whole core should be modeled with a single section.

W3(I) Radial reflector channel number.

W4(R) Starting time of the transient advancement of MASTER (sec).

W5(I) Trip number (reactor trip card number).

17.13.3 CARD 61100002, Control variable numbers for "control rod bank position" control

(Maximum 10 words can be entered)

W1(I) Control variable number for control rod bank 1. The bank position is represented in cm.

W2(I) Control variable number for control rod bank 2. The bank position is represented in cm.

17.13.4 CARD 6110NNXX, Axial Power Tables

This is repeated naxp times

17.13.4.1 CARD 6110NN00

W1(I) i Enter axial power profile table identification number.

W2(I) naxn(i) Enter number of pairs of elements in axial power profile table i.

17.13.4.2 CARD 6110NN01 through 6110NN99

W1(R) y(i,n) Enter vertical location, relative to bottom of the section 1, where axial power factor axial(i,n) is applied. Use inches or meters.

W2(R) axial(i,n) Enter relative power factor (the ratio of local power to average power) at vertical location y(i,n).

All rods using the same table should start and end at the same vertical locations. In the table, y(i,n) must be the vertical location of the beginning of the rods, and y(i,naxn(i)) must be the vertical location of the end of the rods.

17.13.5 CARD 6111XXXX, Power Forcing Function

17.13.5.1 CARD 61110000

The one word has to be entered on this card.

W1(I) nq Enter 1-D general table number for power forcing function .

Note: The power forcing function is still effective when the 3D heat structures use the point kinetics model with "SEPARA3D" module. For a steady-state calculation, users can use this function for regulating the total core power when using the point kinetics model. But, for transient calculations using the point kinetics model, this function should be always 1.0.

17.13.6 CARD 6112XXXX, Gap Conductance Forcing Function

17.13.6.1 CARD 61120000

The one word has to be entered on this card

W1(I) ngpff Enter 1-D general table number for gap conductance forcing function .

17.13.7 CARD 6113xxxx, Direct Moderator Heating

This card must be entered if SEPARA3D option is selected on W2(A) in Card 3000000

17.13.7.1 CARD 61130000

W1(I) ndht_tot Total numeber of nodes for direct heating

W2(R) direct_f Direct heating fraction to total heating power

17.13.7.2 CARD 61130001 through 61130099

W1(I) Channel/node volume number (CCCNN0000)

W2(I) iskp Volume increament

W3(R) qdirect Fraction to total direct heating

W4(R) Numbers of volumes

17.14 CARD 612XXXXX, Turbulent Mixing Data

17.14.1 CARD 61200000

W1(I) n1 Enter number of sections in which turbulence will be applied.

17.14.2 CARD 6120NN00 (NN=1, n1)

Three words have to be entered on each card.

W1(I) i Section index number.

W2(R) beta(i) Mixing coefficient (=w'/G-S)

W3(R) aaak(i) Equilibrium distribution weighting factor in void drift model. Suggested value=1.0.

17.15 CARD 613XXXXX, Boundary Condition Data

17.15.1 CARD 61300000

W1(I) nibnd Enter the total number of vertical mesh cell boundary conditions.

W2(I) nkbnd Enter the total number of transverse momentum cells for which crossflow will be set to zero.

W3(I) nfunct Enter the number of forcing functions for the boundary conditions.

W4(I)	ngbnd	Enter the number of groups of contiguous transverse momentum cells for which crossflows will be set to zero.

Note: The inlet of the lowest channels and the exit of the highest channels in the 3D hydrodynamic input model are automatically blocked. Other boundary conditions specified by users on these cells are effective (overwritten).

In the earlier versions of MARS (e.g., MARS 1.3.1), appropriate boundary conditions should have been provided for all the vertical mesh cells at the inlet of the lowest channels and the exit of the highest channels. Old input data can be run without modification, but it is not efficient.

17.15.2 Forcing Function

17.15.2.1 CARD 61300001 through 61300009 (k=1,nfunct)

Specify Word 3 of 61300000 (nfunct) words. The card numbers range from 61300001 to 61300009 which need not be in sequential order.

W1,2,..,k(I)	npts(k)	Enter the number of points (pairs of values) in each forcing function table.

17.15.2.2 CARD 6130NN00 through 6130NN99 (NN= 1, nfunct)

Specify the data sets corresponding to Word k(npts(k)) of 6130000X. i=1,npts(k)

W1(R)	abscis(nn,i)	Enter the time in seconds, the factor is applied
W2(R)	ordnit(nn,i)	Enter the forcing function factor to be applied at W1 time.
.		
.		
Wnpts(R)	abscis(nn,npts)	
Wnpts+1(R)	ordnit(nn,npts)	

17.15.3 Vertical Mesh Cell Boundary Condition Data

Specify only if W1 of 61300000(nibnd) > 0.

This card is repeated nibnd times.

17.15.3.1 CARD 6131NN00 (NN=1,nibnd)

W1(I) ibound(1,nn) Enter the index number of the channel which boundary condition specified on this card applies to.

W2(I) ibound(2,nn) Enter the vertical node number at which this boundary condition is applied. (NOTE: The node number is referenced to the beginning of the section that the channel identified in W1 resides in.)

W3(I) ispec(nn) Enter the boundary condition type. Valid options are:

1 = pressure and enthalpy boundary condition

2 = flow and enthalpy

3 = flow only

4 = mass source (flow rate and enthalpy)

5 = pressure sink and enthalpy

W4(I) npfn(nn) Enter the index number of the forcing function table by which the first parameter of the boundary condition will be varied. (NOTE: The forcing function tables are numbered sequentially in the order they are read in on card 6130NNXX) For example: If W3(ispec(nn))= 1 and W4(npfn(nn)) = 3, the specified pressure will be adjusted according to the third forcing function entered on 6130NNXX. Enter zero if the boundary condition is constant.

W5(I) nhfn(nn) Enter the index number of the forcing function table by which the second parameter of the boundary condition will be varied. (For example: If W3(ispec(nn)) = 1 and W5(nhfn(nn)) = 6, the specified enthalpy will be adjusted according to the 6th forcing function specified on 6130NNXX.) Enter zero if the boundary condition is constant.

W6(R) pvalue(nn) Enter the first boundary value.

If W3(ispec(nn)) = 1 or 5, enter pressure (psia or N/m^2).

If W3(ispec(nn)) = 2,3 or 4, enter flow rate (lbm/sec or kg/sec).

W7(R) hvalue(nn) Enter enthalpy (Btu/lbm or J/kg). Enter zero (0) if W3(ispec(nn)) = 3.

W8(R) xvalue(nn) Pressure (psia or N/m^2) must be input for W3(ispec(nn)) = 2, 3 or 4.

17.15.3.2 CARD 6131NN10

k = W1(ngas) on 60110000 + 2. Specify only if W3 of 6131NN00(ispec(nn)) ≠ 3.

W1(R) hmga(nn) Enthalpy of noncondensable gas mixture.

W2,3,..,k(R) gvalue(nn) Volume fraction of gas in vapor-gas mixture. (specify in same order as in card 601XXXXX)

Ex) 61310110 H_ncg vf_liq vf_steam vf_gas1 vf_gas2 ..., .

H: Enthalpy, ncg: Non-Condensible Gas, vf: Volume Fraction

17.15.3.3 CARD 6131NN20

W1(I) nhmfn(nn) Index number of forcing function applied to gas mixture enthalpy.

W2,3,..,k(I) ngfn(nn) Index number of forcing function applied to volume fraction of each gas.

17.15.3.4 CARD 6131NN30

Specify only if some W3 of 6131NN00(ispec(nn)) = 4.

W1(R) ainjt(nn) Enter the flow area of the mass injection (in^2 or m^2).

17.15.3.5 CARD 6131NN40

Specify only if some W3 of 6131NN00(ispec(nn)) = 5.

W1(R) asink(nn) Enter the flow area of the pressure sink (in^2 or m^2).

W2(R) sinkk(nn) Enter the loss coefficient (velocity head) of the pressure sink.

W3(R) dxsink(k) Enter the length of the momentum control volume for the sink (inches or meters).

17.15.4 CARD 6132NN00, Gap Boundary Condition Data

17.15.4.1 CARD 6132NN00

This card is read W4 of 61300000(ngbnd) times.

W1(I) k Enter the gap number to which a zero (0.0) crossflow is to be applied.

W2(I) jstart Enter the continuity cell number at which to start applying the zero crossflow.

W3(I) jend Enter the continuity cell number at which to stop applying the zero crossflow.

Note: The crossflow will be set to zero for gap k between nodes W2(jstart) and W3(jend). The node numbers are given relative to the beginning of the section containing gap W1(k).

This card may be repeated as many times as necessary for a given gap W1(k), to identify all axial levels that have zero crossflow. The total number of transverse momentum cells with zero crossflow boundary conditions specified by card 6132NN00 must sum to W2 of 61300000(nkbnd).

17.16 CARD 614XXXXX, Output Option.

17.16.1 CARD 61400000

W1(I) n1 Enter the general vessel output option. Valid entries are:

1 = print channels only

2 = print channels and gaps only

3 = print rods and unheated conductors only

4 = print rods, unheated conductors, and channels only

5 = print channels, gaps, rods, and unheated conductors

W2(I) nout1 Enter the number of channels to be printed (used if W1(n1)=1, 2, 4, or 5).

If W2(nout1) = 0, all channels will be printed.

If W2(nout1) > 0, an array of W2(nout1) channel numbers must be entered on card 614100XX.

W3(I) nout2 Enter the number of rods to be printed (used if W1(n1) > 2).

If W3(nout2) = 0, all rods will be printed.

If W3(nout2) > 0, an array of W3(nout2) rod numbers must be entered on card 614200XX.

W4(I) nout3 Enter the number of gaps to be printed (used if W1(n1) = 2 or 5).

If W4(nout3) = 0, all gaps will be printed.

If W4(nout3) > 0, an array of W4(nout3) gap numbers must be entered on card 614300XX.

W5(I) nout4 Enter the number of unheated conductors to be printed (used if W1(n1) > 2).

If W5(nout4) = 0, all unheated conductors will be printed.

If W5(nout4) > 0, an array of W5(nout4) unheated conductor numbers must be entered on card 614400XX.

W6(I) ipropp Enter the property table print option.

Valid entries are:

0 = do not print the property table

1 = print the property table

W7(I) iopt Enter the debug print option. Valid entries are:

0 = normal printout only

2 = debug printout (print extra data for channels, rods and gaps)

17.16.2 CARD 6141000N, Channel Option.

Specify only if W1 of 61400000(n1) is not 3 and W2 of 61400000(nout1) is greater than 0.

W1,2,..,k(I) ptintc(i) Enter the index numbers of channels to be printed.

Repeat this card until nout1 values have been entered.

17.16.3 CARD 6142000N, Gap Option.

Specify only if W1 of 61400000(n1) is not 3 and W4 of 61400000(nout3) is greater than 0.

W1,2,..,k(I) ptintg(i) Enter the index numbers of gaps to be printed.

Repeat this card until nout3 values have been entered.

17.16.4 CARD 6143000N, Rod Option.

Specify only if W1 of 61400000(n1) is not 3 and W3 of 61400000(nout2) is greater than 0.

W1,2,..,k(I)	ptintr(i)	Enter the index numbers of rods to be printed.

Repeat this card until nout2 values have been entered.

17.16.5 CARD 6144000X, Unheated Conductor Option.

Specify only W5 of 61400000(n1) is greater than 0.

W1,2,..,k(I)	ptints(i)	Enter the index numbers of unheated conductors to be printed.

Repeat this card until nout5 values have been entered.

17.17 CARD 615XXXXX, Graphic Option.

17.17.1 CARD 61500000

This card is optional for all problem types. If the card is omitted, default values are assumed (W1=1000, W2=0, and W3=0).

W1(I)	mxgdmp	Enter the maximum number of time steps for which graphics data will be saved. Absolute maximum is 1000. Note: This cannot be changed on a restart.
W2(I)	igrfop	Enter zero (0).
W3(I)	nllr	Enter the number of liquid level calculations. (Number of 61500100 cards.) (Valid only if W2 > 0.)

17.18 CARD 616XXXXX, Initial Condition Data.

These cards can be specified if W2 of card 60000000 is 2.

17.18.1 CARD 61600NNN, INITIAL VOID FRACTIONS

The three words are entered on this card, where NNN =1, 2, 3,..., (Total number of the 3D cells).

W1(I)	i	Enter the channel number (sequential number).
W2(I)	j	Enter the mesh number, where j=2,3,...,jnodes+1.

W3(R)	al	Enter the void fraction.